Ex no: 5
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Experiment on packet capture tool : Wireshark.

Aim:

To understand the feature of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

Packet Sniffer:

- Captures packets sent and received by the computer.
- Displays protocol information of captured packets.
- Works passively.
- Receives copies of Network packets for Analysis.

Packet Sniffer Tools:

- TCP dump - Command line packet capture tool.
- Wireshark - Graphical packet analyser.

Description

Wireshark is a network protocol analyzer that captures and displays network packets in real time. It helps monitor, analyze, troubleshoot and inspect network traffic using filters and packet details

Use of Networks:

- Capture network traffic.
- Analyzers protocols.
- Applies displays filters
- monitors Network issues

Applications:

- Network troubleshooting.
- Security analysis
- Protocol debugging
- Learning network protocols.

capturing packet:

1. Open Wireshark.
2. Select the network interface
3. Click start capture.
4. Observes packet in real time.

Wireshark panes:

- packet list: Display capture packets
- packet Details: Shows protocols information.
- packet Bytes: Display packet data in hexadecimal notation

color coding

- purple: TCP
- Blue: UDP
- Black: Error packets.

Filtering packets:

- Enter a letter in the filter bar.
- Right click to apply filters or follows the TCP stream.

Flow Graph:

Display the communication flow between devices, making packing exchange easier to understand.

Capture 100 packets.

1. Open Wireshark.
2. Select the network interface.
3. Set the capture limit into 100 packets.
4. start capturing
5. Stop and save the captured packets.

Analysing packets using wireshark tool.

Create a filter to Display only TCP/UDP packets.

1. Open the capture file.
2. Apply the TCP || Udp filter.
3. Inspect and view the flow graph.

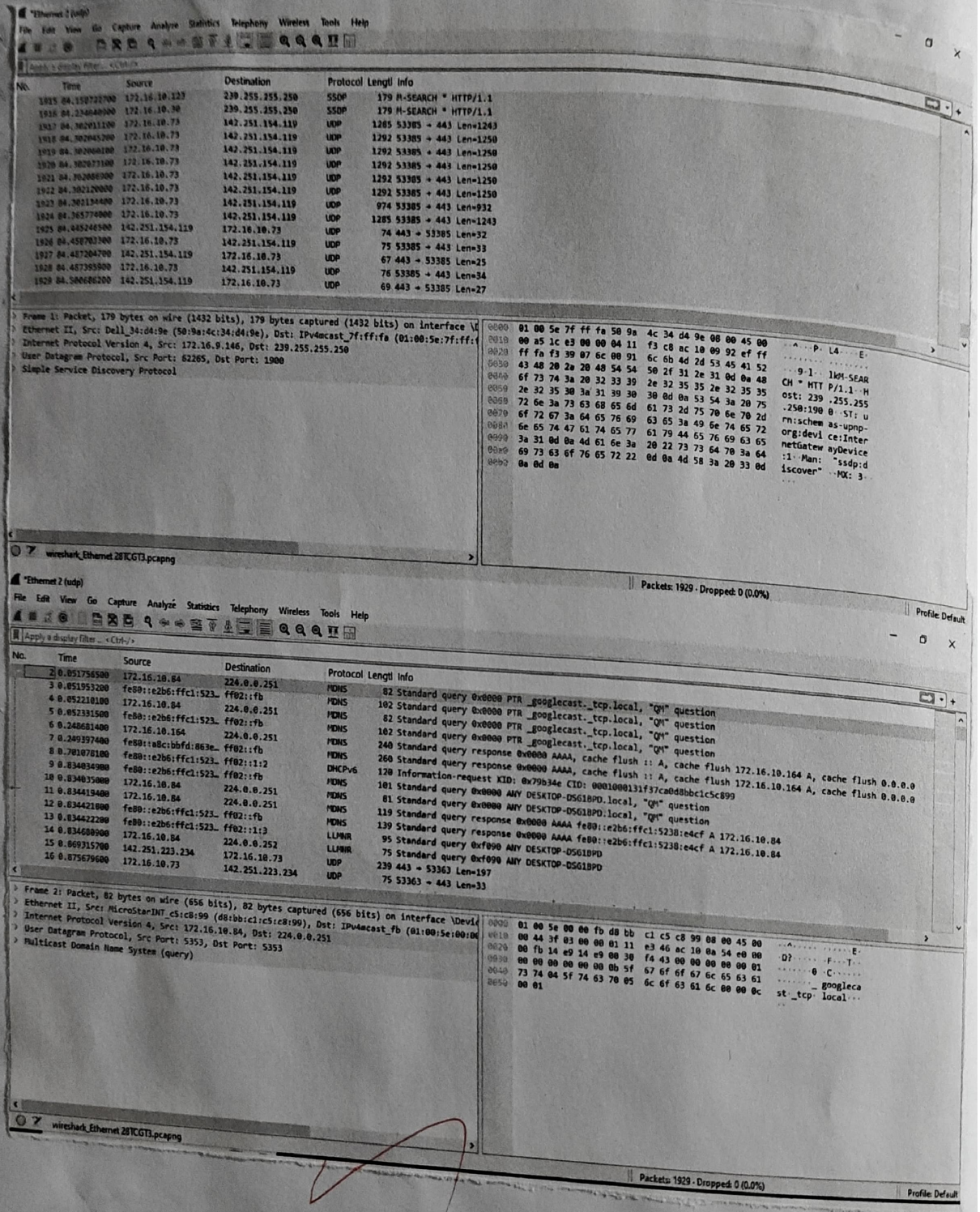
2. Create a file to display only ARP packets.
   1. Open the capture file.
   2. Apply the Carp filter.
   3. Inspect and view the flow graph.
3. Create a filter to display only DNS packets.
   1. Open the capture file
   2. Apply the DNS filter.
   3. Inspect and view the flow graph.
4. Create a filter to display only IP/ICMP packets
   1. Open the capture file.
   2. Apply the ip & icmp filter.
   3. Inspect and view the flow graph and save capture.

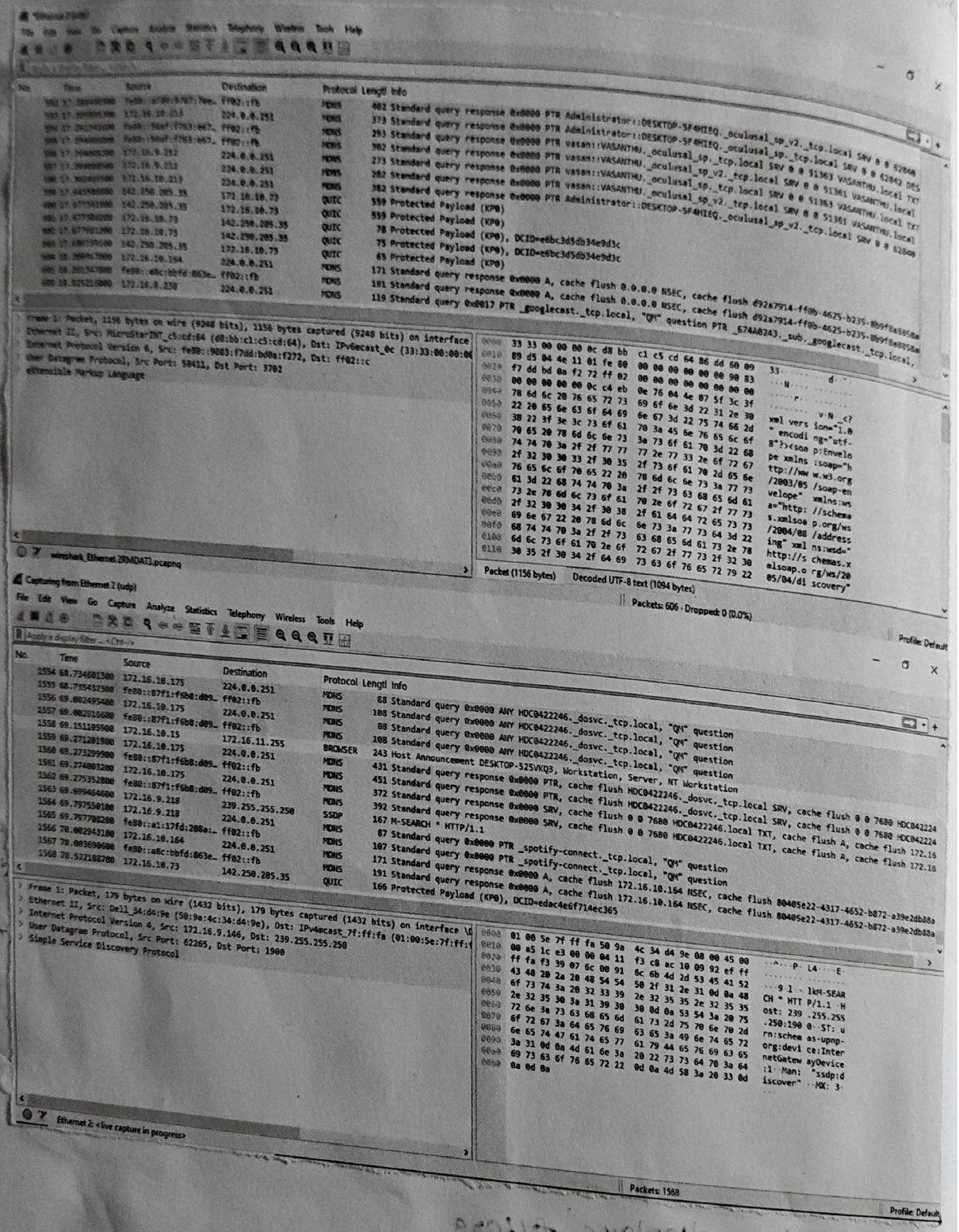
5) Create a filter to display only DHCP packets

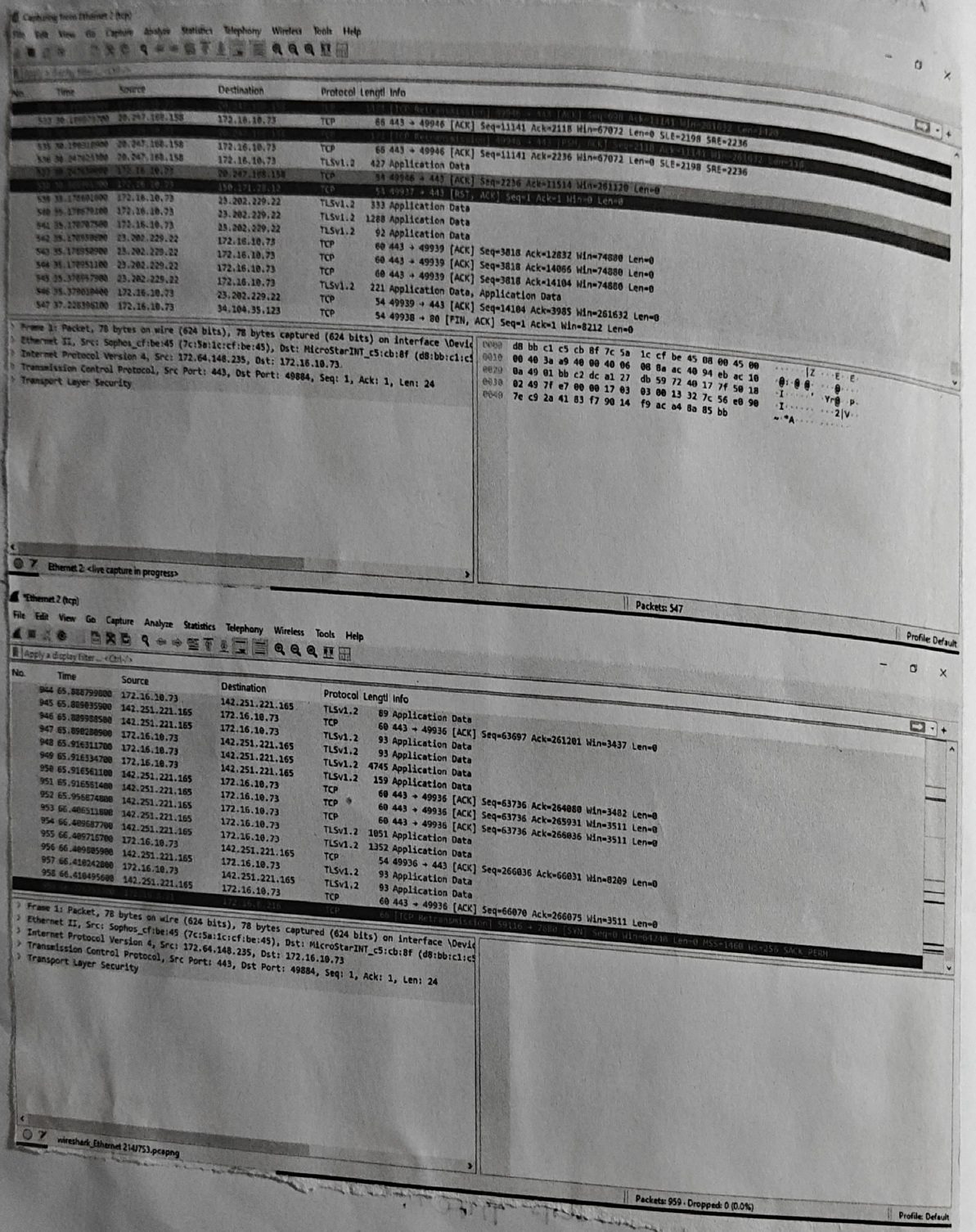
1. Open the capture file
2. Apply the dhcp file.
3. Inspect and view the flow graph and save the capture.

Result:

Thus the feature of wireshark and the implement of different filter was successfully compiled.


*Ethernet 2 (udp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
1915 84.150722700 172.16.10.123 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
1916 84.234848000 172.16.10.30 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
1917 84.302911100 172.16.10.73 142.251.154.119 UDP 1285 53385 → 443 Len=1243
1918 84.302945200 172.16.10.73 142.251.154.119 UDP 1292 53385 → 443 Len=1250
1919 84.302960100 172.16.10.73 142.251.154.119 UDP 1292 53385 → 443 Len=1250
1920 84.302973100 172.16.10.73 142.251.154.119 UDP 1292 53385 → 443 Len=1250
1921 84.302986900 172.16.10.73 142.251.154.119 UDP 1292 53385 → 443 Len=1250
1922 84.303120000 172.16.10.73 142.251.154.119 UDP 1292 53385 → 443 Len=1250
1923 84.303134400 172.16.10.73 142.251.154.119 UDP 974 53385 → 443 Len=932
1924 84.365770000 172.16.10.73 142.251.154.119 UDP 1285 53385 → 443 Len=1243
1925 84.445246500 142.251.154.119 172.16.10.73 UDP 74 443 → 53385 Len=32
1926 84.458703300 172.16.10.73 142.251.154.119 UDP 75 53385 → 443 Len=33
1927 84.487204700 142.251.154.119 172.16.10.73 UDP 67 443 → 53385 Len=25
1928 84.487395900 172.16.10.73 142.251.154.119 UDP 76 53385 → 443 Len=34
1929 84.500686200 142.251.154.119 172.16.10.73 UDP 69 443 → 53385 Len=27
Frame 1: Packet, 179 bytes on wire (1432 bits), 179 bytes captured (1432 bits) on interface \D
Ethernet II, Src: Dell_34:d4:9e (50:9a:4c:34:d4:9e), Dst: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:
Internet Protocol Version 4, Src: 172.16.9.146, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 62265, Dst Port: 1900
Simple Service Discovery Protocol
0000 01 00 5e 7f ff fa 50 9a 4c 34 d4 9e 08 00 45 00 ..^...P. L4....E.
0010 00 a5 1c e3 00 00 04 11 f3 c8 ac 10 09 92 ef ff ....
0020 ff fa f3 39 07 6c 00 91 6c 6b 4d 2d 53 45 41 52 ...9.l.. lkM-SEAR
0030 43 48 20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 CH * HTT P/1.1..H
0040 6f 73 74 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 ost: 239 .255.255
0050 2e 32 35 30 3a 31 39 30 30 0d 0a 53 54 3a 20 75 .250:190 0..ST: u
0060 72 6e 3a 73 63 68 65 6d 61 73 2d 75 70 6e 70 2d rn:schem as-upnp-
0070 6f 72 67 3a 64 65 76 69 63 65 3a 49 6e 74 65 72 org:devi ce:Inter
0080 6e 65 74 47 61 74 65 77 61 79 44 65 76 69 63 65 netGatew ayDevice
0090 3a 31 0d 0a 4d 61 6e 3a 20 22 73 73 64 70 3a 64 :1..Man: "ssdp:d
00a0 69 73 63 6f 76 65 72 22 0d 0a 4d 58 3a 20 33 0d iscover" ..MX: 3.
00b0 0a 0d 0a ...
wireshark_Ethernet 28TCGT3.pcapng
Packets: 1929 · Dropped: 0 (0.0%)
Profile: Default
*Ethernet 2 (udp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
2 0.051756500 172.16.10.84 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
3 0.051953200 fe80::e2b6:ffc1:523… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
4 0.052210100 172.16.10.84 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
5 0.052331500 fe80::e2b6:ffc1:523… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
6 0.248681400 172.16.10.164 224.0.0.251 MDNS 240 Standard query response 0x0000 AAAA, cache flush :: A, cache flush 172.16.10.164 A, cache flush 0.0.0.0
7 0.249397400 fe80::a8c:bbfd:863e… ff02::fb MDNS 260 Standard query response 0x0000 AAAA, cache flush :: A, cache flush 172.16.10.164 A, cache flush 0.0.0.0
8 0.701078100 fe80::e2b6:ffc1:523… ff02::1:2 DHCPv6 120 Information-request XID: 0x79b34e CID: 000100013 1f37ca0d8bbc1c5c899
9 0.834034900 fe80::e2b6:ffc1:523… ff02::fb MDNS 101 Standard query 0x0000 ANY DESKTOP-DSG1BPD.local, "QM" question
10 0.834035000 172.16.10.84 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-DSG1BPD.local, "QM" question
11 0.834419400 172.16.10.84 224.0.0.251 MDNS 119 Standard query response 0x0000 AAAA fe80::e2b6:ffc1:5238:e4cf A 172.16.10.84
12 0.834421800 fe80::e2b6:ffc1:523… ff02::fb MDNS 139 Standard query response 0x0000 AAAA fe80::e2b6:ffc1:5238:e4cf A 172.16.10.84
13 0.834422200 fe80::e2b6:ffc1:523… ff02::1:3 LLMNR 95 Standard query 0xf090 ANY DESKTOP-DSG1BPD
14 0.834600900 172.16.10.84 224.0.0.252 LLMNR 75 Standard query 0xf090 ANY DESKTOP-DSG1BPD
15 0.869315700 142.251.223.234 172.16.10.73 UDP 239 443 → 53363 Len=197
16 0.875679600 172.16.10.73 142.251.223.234 UDP 75 53363 → 443 Len=33
Frame 2: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Devi
Ethernet II, Src: MicroStarINT_c5:c8:99 (d8:bb:c1:c5:c8:99), Dst: IPv4mcast_fb (01:00:5e:00:0
Internet Protocol Version 4, Src: 172.16.10.84, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
0000 01 00 5e 00 00 fb d8 bb c1 c5 c8 99 08 00 45 00 ..^..... ......E.
0010 00 44 3f 03 00 00 01 11 e3 46 ac 10 0a 54 e0 00 .D?..... .F...T..
0020 00 fb 14 e9 14 e9 00 30 f4 43 00 00 00 00 00 01 .... 0.C.....
0030 00 00 00 00 00 00 0b 5f 67 6f 6f 67 6c 65 63 61 .... _googleca
0040 73 74 04 5f 74 63 70 05 6c 6f 63 61 6c 00 00 0c st._tcp. local...
0050 00 01 ..
wireshark_Ethernet 28TCGT3.pcapng
Packets: 1929 · Dropped: 0 (0.0%)
Profile: Default




Capturing from Ethernet 2 (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
172.16.10.73 TCP 68 443 → 49946 [ACK] Seq=11141 Ack=2118 Win=67072 Len=0 SLE=2198 SRE=2236
172.16.10.73 TLSv1.2 427 Application Data
20.247.168.158 TCP 54 49946 → 443 [ACK] Seq=2236 Ack=11514 Win=261120 Len=0
150.171.28.12 TCP 54 49937 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
172.16.10.73 23.202.229.22 TLSv1.2 333 Application Data
23.202.229.22 TLSv1.2 1288 Application Data
172.16.10.73 TLSv1.2 92 Application Data
23.202.229.22 TCP 60 443 → 49939 [ACK] Seq=3818 Ack=12832 Win=74880 Len=0
172.16.10.73 TCP 60 443 → 49939 [ACK] Seq=3818 Ack=14066 Win=74880 Len=0
23.202.229.22 TCP 60 443 → 49939 [ACK] Seq=3818 Ack=14104 Win=74880 Len=0
172.16.10.73 TLSv1.2 221 Application Data, Application Data
23.202.229.22 TCP 54 49939 → 443 [ACK] Seq=14104 Ack=3985 Win=261632 Len=0
172.16.10.73 34.104.35.123 TCP 54 49938 → 80 [FIN, ACK] Seq=1 Ack=1 Win=8212 Len=0
Frame 1: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Devic
Ethernet II, Src: Sophos_cf:be:45 (7c:5a:1c:cf:be:45), Dst: MicroStarINT_c5:cb:8f (d8:bb:c1:c5
Internet Protocol Version 4, Src: 172.64.148.235, Dst: 172.16.10.73
Transmission Control Protocol, Src Port: 443, Dst Port: 49884, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
Ethernet 2: <live capture in progress>
Packets: 547
Profile: Default
Capturing from Ethernet 2 (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
944 65.888799800 172.16.10.73 142.251.221.165 TLSv1.2 89 Application Data
945 65.889035900 142.251.221.165 172.16.10.73 TCP 60 443 → 49936 [ACK] Seq=63697 Ack=261201 Win=3437 Len=0
946 65.889988500 142.251.221.165 172.16.10.73 TLSv1.2 93 Application Data
947 65.898280900 172.16.10.73 142.251.221.165 TLSv1.2 93 Application Data
948 65.916311700 172.16.10.73 142.251.221.165 TLSv1.2 4745 Application Data
949 65.916334700 172.16.10.73 142.251.221.165 TLSv1.2 159 Application Data
950 65.916561100 142.251.221.165 172.16.10.73 TCP 60 443 → 49936 [ACK] Seq=63736 Ack=264080 Win=3482 Len=0
951 65.916561400 142.251.221.165 172.16.10.73 TCP 60 443 → 49936 [ACK] Seq=63736 Ack=265931 Win=3511 Len=0
952 65.956874800 142.251.221.165 172.16.10.73 TCP 60 443 → 49936 [ACK] Seq=63736 Ack=266036 Win=3511 Len=0
953 66.406511800 142.251.221.165 172.16.10.73 TLSv1.2 1051 Application Data
954 66.409687700 142.251.221.165 172.16.10.73 TCP 54 49936 → 443 [ACK] Seq=266036 Ack=66031 Win=8209 Len=0
955 66.409716700 172.16.10.73 142.251.221.165 TLSv1.2 1352 Application Data
956 66.409885900 142.251.221.165 172.16.10.73 TLSv1.2 93 Application Data
957 66.410242800 172.16.10.73 142.251.221.165 TCP 60 443 → 49936 [ACK] Seq=66070 Ack=266075 Win=3511 Len=0
958 66.410495600 142.251.221.165 172.16.10.73
Frame 1: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Devic
Ethernet II, Src: Sophos_cf:be:45 (7c:5a:1c:cf:be:45), Dst: MicroStarINT_c5:cb:8f (d8:bb:c1:c5
Internet Protocol Version 4, Src: 172.64.148.235, Dst: 172.16.10.73
Transmission Control Protocol, Src Port: 443, Dst Port: 49884, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
wireshark_Ethernet 2I4I753.pcapng
Packets: 959 · Dropped: 0 (0.0%)
Profile: Default